UROP · Text + image · Part 2

Execution handoff · MM-PART2-v2 · 23 October 2026

Status: PLAN ONLY. Part 1 has not yet run. Use GPT-6.1 Sol with High reasoning effort. Execute this part after the Part 1 receipt passes its acceptance checks. Extra High is optional for a specific hard recovery/debugging problem; no assistant model guarantees higher classifier accuracy.

01 · Objective and prerequisites

Complete the same OFFICIAL-6W-MM-v1 seed42 BERT-base + ResNet-50 maximum-fusion run begun in Part 1. Resume epochs 3–4, or satisfy the registered early-stop rule; select by full-validation Macro-F1; evaluate the frozen selected checkpoint once on the public test; verify and archive all evidence and update GitHub.

Work locally only in `/Users/panveenaidu/Downloads/SEM7/UROP`. Preserve original TSVs, old protocols, historical subset/pilot, completed text-only results, and unrelated user edits. Synced ChatGPT project `sources/` files are read-only.

Read these before acting:

1. `context/MULTIMODAL_PART1_EXECUTION_PLAN_20261023.md` in full; its dataset, architecture, optimizer, recovery, storage and reserve requirements remain binding.

2. `context/RESUME_HERE.md`, `PROJECT_CONTEXT.md`, `DECISIONS.md`, `EXPERIMENT_LOG.md`, `TASK_REPORT.md`, `collab/COLAB_WORKLOG.md`, `work_logs/MODEL_STATUS.md`, `work_logs/TROUBLESHOOTING.md`, `README.md`, `results/PROGRESS_TRACKER.json`.

3. The actual `results/MM_PART1_RECEIPT_<date>.json`, Part 1 report, run protocol/source/config, coverage and warm-start receipts, checkpoint manifest and Git receipt. Resolve exact paths from those receipts; do not invent filenames or assume a planned file exists.

4. `results/TEXT_ONLY_FINAL_REPORT_20261009.md`, official dataset provenance/audit, the faculty dataset, paper Tables 4–6, authors' instructions, and the local literature survey for comparison limitations.

Part 1 must prove all 682,661 images usable; 564,000 training pairs and 59,342 validation pairs processed in each of two epochs; verified next-epoch recovery; durable local/private cloud artifacts; updated context and verified Git backup; no final multimodal test. If any gate is incomplete, report it and perform bounded integrity checks. Do not call Part 1 complete or silently start a different run. Ask for a scope decision if finishing missing Part 1 work would require changing cohort or protocol.

02 · Browser, account and cloud preparation

Use **Chrome profile Rocky, panvee62@gmail.com**, with the existing authorized private Kaggle workspace. Verify the actual account/workspace before writing or launching. Native computer/desktop controls may open or focus Chrome and the dedicated profile; then prefer enabled Chrome extension/browser controls. A separate signed-out automation browser is not the user's authenticated Rocky session. Return to the correct profile instead of migrating accounts.

If page control fails, preserve evidence and restore the extension connection. Hand genuine private sign-in/consent to the user. Do not use Colab CLI/new OAuth, buy compute, switch accounts or create a scheduled task. Colab through Chrome is fallback only after confirming compatible runtime and recoverable inputs.

Check live assistant allowance, current GPU allowance, active jobs, actual device, session duration, disk/input/output limits and saved-version persistence. Old T4 allocation and 29h09m quota are historical. Prefer the previously verified single-GPU path; do not assume T4 x2 means two GPUs were used or introduce distributed training mid-run.

Load the exact private input version, complete paired manifest and source/config. Recheck all hashes in the runtime. Confirm no duplicate trainer exists. Reserve room for two recovery generations, selected weights, both evaluation exports and archives. Keep image inputs separate from retained output.

03 · Restore and finish training

1. Verify Part 1's checkpoint bytes/hash and corresponding generation metadata. Strict-load model including BatchNorm buffers, optimizer, four-epoch scheduler, AMP scaler, all RNG, sampler order/cursor, deterministic crop state, selection/early-stop history and source/config/data hashes.

2. Use the verified tensor's consumed position, not newer UI/log text. Confirm epoch 3's next IDs, crop parameters, learning rate and update counters match the expected continuation. If epoch 2 validation alone is unfinished, resume that validation without retraining the epoch. An earlier reliable tensor requires explicitly recorded replay, not skipping unsaved work.

3. Keep registered settings: max four epochs total, microbatch8/accumulation4, eval batch32, max128 tokens, BERT lr1e-5/image1e-4/head1e-3 with the existing schedule. Do not reset optimizer/scheduler or use a new warm start. Train both encoders/head; preserve the original preprocessing and effective batch.

4. Complete epochs 3–4 with all 564,000 train pairs and all 59,342 validation pairs per epoch, or apply the exact registered early-stop rule (earliest epoch3, patience2, delta .001). Keep selection based on highest validation Macro-F1, then lower loss, then earlier epoch.

5. Checkpoint at cleared-gradient boundaries and each completed epoch, preserve two generations, and verify durable private outputs between bounded jobs. Use Part 1's measured throughput/export times to fit jobs with 20% margin and at least one hour before the actual session cap. Split jobs if necessary; do not shrink data or alter budget.

6. Inspect validation losses and class-wise errors. A fixed validation-only image-shuffle diagnostic may measure image sensitivity if preregistered and affordable; it is not an image-only baseline or permission to redesign this run. If the run appears undertrained or misses the target, record it and propose a separate future experiment.

Timing: replace old estimates with the measured Part 1 throughput. Rough planning is another 8–12+ GPU hours for two epochs/validation, plus final selected-checkpoint evaluation, transfers and verification. This is conditional, not a deadline promise. Check remaining assistant allowance before every new cloud job.

04 · Freeze selection and perform the final test

After training and full validation are complete, freeze the selected epoch, weight hash, model/config/source/data hashes and selection receipt. Report accuracy and all other metrics from that SAME selected checkpoint. Do not switch to whichever checkpoint has the best test accuracy.

Run full selected-checkpoint validation if its required exports are missing, then evaluate **all 59,319 public-test rows once**. Integrity checks on test images are permitted earlier, but model selection and tuning must use training/validation only. Keep a test-start and test-completion receipt. If inference is interrupted, complete or explicitly replay the same frozen evaluation and log it; do not treat retries as new models or use partial test metrics to tune.

Export for validation and test:

• Exact ID, split, gold label, predicted label, all six probabilities and per-row cross-entropy loss.

• Accuracy, micro-F1, Macro-F1, weighted-F1, balanced accuracy, sample-mean loss.

• Per-class precision/recall/F1/support and both raw and row-normalized 6x6 confusion matrices.

• Selected epoch/hash, sample counts, software versions, device/used-device count, throughput, training/inference/export time, peak GPU memory, parameter counts, successful/skipped updates and total compute including prior text warm start.

Independently verify exact official ID/label sets, no omissions/duplicates, finite normalized probabilities, prediction consistency and confusion/support totals. Recompute metrics from predictions and mean loss from per-row CE; maximum confidence alone cannot reconstruct CE. Load saved selected weights and perform a small roundtrip check. Archive checksums/CRCs and required members must pass before completion.

05 · Compare honestly and report

Six-way multimodal accuracy	Validation	Test
Paper BERT + ResNet-50 maximum fusion	86.00%	85.88%
Minimum aspiration: +1 percentage point	87.00%	86.88%
Our completed run	Fill verified value	Fill verified value

Source: paper Tables 4–5. The older +2–3-point ambition remains a stretch goal. Record each split's delta in percentage points and whether its target was met. No fabricated/estimated final scores or test-driven redesign.

The completed text-only result is validation82.7407%/test82.3901% accuracy, with Macro-F1 77.8682%/77.4566%. It used all released rows and is a reference, not a prediction for multimodal performance. Paper comparisons are single-seed numerical comparisons with different feature/fine-tuning pipelines and an unresolved 335-row paper/release count difference. Do not claim exact reproduction, significance or state of the art.

The paper does not give comparable macro/weighted-F1, balanced accuracy or loss; show “not reported.” Table6 class-error proportions are not class F1. Micro-F1 equals accuracy for this exhaustive single-label six-class setup, but keep both named as requested.

Write `results/MULTIMODAL_FINAL_REPORT_<date>.md` and a compact offline faculty HTML with the actual connected architecture, workflow, clickable faculty dataset, all validation/test metrics, six-class results, paper comparisons and one honest outcome sentence. Preserve the existing text HTML. If the target is missed, retain the verified baseline and recommend a separately registered validation-led improvement; no extra models are part of this task.

06 · Usage reserve, durable saving and GitHub

Check applicable short/weekly assistant limits before work/jobs and at checkpoints; use the smaller remaining allowance. Check periodically during active work, around every ten minutes when supported. GPU quota is independent. If tools cannot expose current assistant usage, state that and use bounded stages rather than inventing percentages.

Remaining allowance	Required response
Around25%	Synchronize early; prepare a bounded stop and handoff.
At/below20%	No new job/large transfer/refactor. Reach safe checkpoint, export/retrieve/verify, update all records and Git.
At/below10%	Substantive work stopped; only essential verification/receipt/handoff.
At/below5%	Hard stop assistant work; disclose exact saved state and anything unverified. Saving should already be complete.

Apply the lowest crossed threshold if allowance drops suddenly. The 5% reserve is not a planned start for saving. Use graceful checkpoint/stop/export, not a kill mid-write. Scratch checkpoints are not durable cloud output; verify a saved private version or local retrieval before stopping the runtime. A submitted bounded job may continue independently only when persistence is verified and that unsupervised state is disclosed. Never promise inactive-agent monitoring or automatic local/Git updates.

Update `context/PROJECT_CONTEXT.md`, `RESUME_HERE.md`, `DECISIONS.md`, `EXPERIMENT_LOG.md`, `TASK_REPORT.md`, `collab/COLAB_WORKLOG.md`, `work_logs/MODEL_STATUS.md`, README and `results/PROGRESS_TRACKER.{json,md}` at durable boundaries. Record dates, attempt/stage, exact artifact paths/counts/hashes, cloud version/URL, errors, observed quotas and next action. Update troubleshooting for new failures.

Keep raw TSVs, images, weights and ZIPs ignored in `data/` and private cloud. Stage only reviewed source/config/notebooks, sanitized small metrics/reports/manifests/context. Large predictions stay private with a Git-backed manifest/hash. Preserve unrelated changes. Commit/push normally, verify remote commit and write a receipt. If push fails, mark backup pending. Never claim all data/weights are in GitHub.

07 · Completion and next research tasks

Part 2 is complete only when the registered training budget/early stop is satisfied, validation selection is frozen, full test is verified, selected weights and recovery/evidence are archived/reloadable, report/HTML and context are updated, and Git backup is verified. Re-invocation must detect verified completion and avoid training/test reruns.

Workflow checkpoint	Expected when Part 2 accepted
Text-only	100% preserved
Full text+image training	100%
Full text+image validation	100%
Full text+image test + archive	100%
Full text+image workflow	100%
Image-only	20% unchanged
Three-modality overall	73%

These workflow percentages do not depend on beating the paper. Missing image-only work means the whole research project is not 100%. Future work: matched image-only ResNet50 control, repeated seeds and confidence estimates, then one separately registered modern/alignment improvement if justified by validation and compute. Do not automatically launch these.

08 · Paste this to the execution agent

Execute UROP text+image Part 2 using `/Users/panveenaidu/Downloads/SEM7/UROP/context/MULTIMODAL_PART2_EXECUTION_PLAN_20261023.md` (MM-PART2-v2). Use GPT-6.1 Sol High. Read this file and the full Part1 execution plan, then verify the actual Part1 receipt and artifacts. Create or reuse a Goal with this part's acceptance criteria. Work locally only inside UROP. Use desktop controls to open/focus Chrome profile Rocky (panvee62@gmail.com), then the enabled Chrome extension/browser controls for the existing private Kaggle workspace. Verify account and saved inputs before actions.

Resume the same registered four-epoch BERT-base + ResNet50 run; complete epochs3–4 or its legitimate early stop, preserving optimizer/scheduler/AMP/RNG/order. Select by full-validation Macro-F1, freeze selected hashes, evaluate the full public test once, and independently verify all metrics/class results/predictions and archives. Aim for at least87.00% validation and86.88% test accuracy without guaranteeing it or tuning on test. Produce the final report/offline faculty HTML, update all context/log/status/progress files, safely push appropriate files and verify GitHub backup.

At20% allowance enter checkpoint-and-handoff reserve; at10% stop substantive work; at5% hard-stop assistant work with the saved status. Pause the Goal after a verified reserve handoff if incomplete; mark complete only when all acceptance checks pass. If prerequisites are missing, report exact evidence and do not silently restart or change the experiment. No extra models, paid compute, account migration, new scheduler or Colab CLI. Finish with workflow bars, full metrics, target comparison, artifact locations/hashes, cloud version and Git receipt.

This is an ordinary reusable prompt, not an installed slash command. Scientific scope changes or genuine private login/consent may require the user; routine authorized steps should proceed without repeated confirmation.